A LOCALLY CLOSED IMAGE OF A MORPHISM OF GROUPOIDS OVER A FIELD IS CLOSED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The morphism of groupoids over a field situation fixing R_1, R_2, a and the structure maps (more-groupoids chapter, Situation situation-morphism-groupoids-on-field, canonical tag 04Q5; kind reference). cited at statement lines 1816-1817. Supplies the notation and assumptions of the statement.
- Restriction of a groupoid over a field to a field extension, with its structural diagram (more-groupoids chapter, Lemma lemma-restrict-groupoid-on-field, canonical tag 04LQ). cited at proof lines 1824-1829. The diagram in its statement exhibits the restrictions as base changes of the universal homeomorphism U' to U, whence the universal homeomorphism property of R_i' to R_i.
- Over a perfect field the closure of the image of a morphism of groupoids is a groupoid scheme (more-groupoids chapter, Lemma lemma-map-groupoids-on-perfect-field-image, canonical tag 04Q8; also in this lane's queue). cited at proof lines 1834-1839. Provides both groupoid structures used in the final step: Z as a groupoid scheme and (R_2)_red as a groupoid scheme.
- An open image of a morphism of groupoids over a field is closed (more-groupoids chapter, Lemma lemma-open-image-is-closed, canonical tag 04Q6; collected on this lane as L22517). cited at proof lines 1840-1844. Applied to a restricted to (R_2)_red as a morphism (R_2)_red to Z: the image, being locally closed and dense in Z, is open in Z, hence closed in Z and equal to Z.

Source reference. Original reference 04Q5: sources/stacks-chow/tags.txt, source lines 6138–6138. Cited at statement lines 1816-1817: notation and assumptions fixing R_1, R_2, a and the structure maps.

Source reference. Original lemma 04LQ: sources/stacks-chow/tags.txt, source lines 6016–6016. Cited at proof lines 1824-1829: the diagram in its statement exhibits the restrictions as base changes of the universal homeomorphism U' to U.

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Source reference. Original lemma 04Q8: sources/stacks-chow/tags.txt, source lines 6141–6141. Cited at proof lines 1834-1839: Z is a groupoid scheme and (R_2)_red is a groupoid scheme over the perfect field.

Source reference. Original lemma 04Q6: sources/stacks-chow/tags.txt, source lines 6139–6139. Cited at proof lines 1840-1844: the image of the restriction (R_2)_red to Z, open in Z, is closed in Z, hence equals Z.

Lemma 0.2. *Notation and assumptions as in Situation 04Q5. If the image $a(R_1)$ is a locally closed subset of R_2 then it is a closed subset.*

Proof. Let $k \subset k'$ be a perfect closure of the field k. Let R_i' be the restriction of R_i via the morphism $U' = \mathrm{Spec}(k') \to \mathrm{Spec}(k)$. Note that the morphisms $R_i' \to R_i$ are universal homeomorphisms as compositions of base changes of the universal homeomorphism $U' \to U$ (see diagram in statement of Lemma 04LQ). Hence it suffices to prove that $a'(R_1')$ is closed in R_2'. In other words, we may assume that k is perfect.

If k is perfect, then the closure of the image is a groupoid scheme $Z \subset R_2$, by Lemma 04Q8. By the same lemma applied to $\mathrm{id}_{R_1} : R_1 \to R_1$ we see that $(R_2)_{red}$ is a groupoid scheme. Thus we may apply Lemma 04Q6 to the morphism $a|_{(R_2)_{red}} : (R_2)_{red} \to Z$ to conclude that Z equals the image of a. $\square$